

Fundamental Calculus/Differential and Integral Calculus

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Reference : Calculus 10-th Edition by Howard Anton,
Irl Bivens and Stephen Davis

Slide-2

- Function
- Chapter:0
- Section:0.1
- Page-25

Representation of function

Four common methods for representing functions are:

- Numerically by tables
- Algebraically by formulas
- Geometrically by graphs
- Verbally

Function (section:0.1-Page-25)

- **0.1.1 definition** If a variable y (*Grade*) depends on a variable x (*writing*) in such a way that each value of x determines exactly one value of y , then we say that y (*Grade*) is a function of x (*writing*).
- INDEPENDENT AND DEPENDENT VARIABLES
- Independent variable x
- Dependent variable y

Function

► **Example 2** The equation

$$y = 3x^2 - 4x + 2$$

has the form $y = f(x)$ in which the function f is given by the formula

$$f(x) = 3x^2 - 4x + 2$$

Find $f(0)$, $f(-1.7)$, $f(\sqrt{2})$

For each input x , the corresponding output y is obtained by substituting x in this formula.
For example,

$$f(0) = 3(0)^2 - 4(0) + 2 = 2$$

f associates $y = 2$ with $x = 0$.

$$f(-1.7) = 3(-1.7)^2 - 4(-1.7) + 2 = 17.47$$

f associates $y = 17.47$ with $x = -1.7$.

$$f(\sqrt{2}) = 3(\sqrt{2})^2 - 4\sqrt{2} + 2 = 8 - 4\sqrt{2}$$

f associates $y = 8 - 4\sqrt{2}$ with $x = \sqrt{2}$.



Domain and Range of function

- Example:

$$y(\text{Grade}) = f(x) = f(\text{performance}).$$

- If x and y are related by the equation

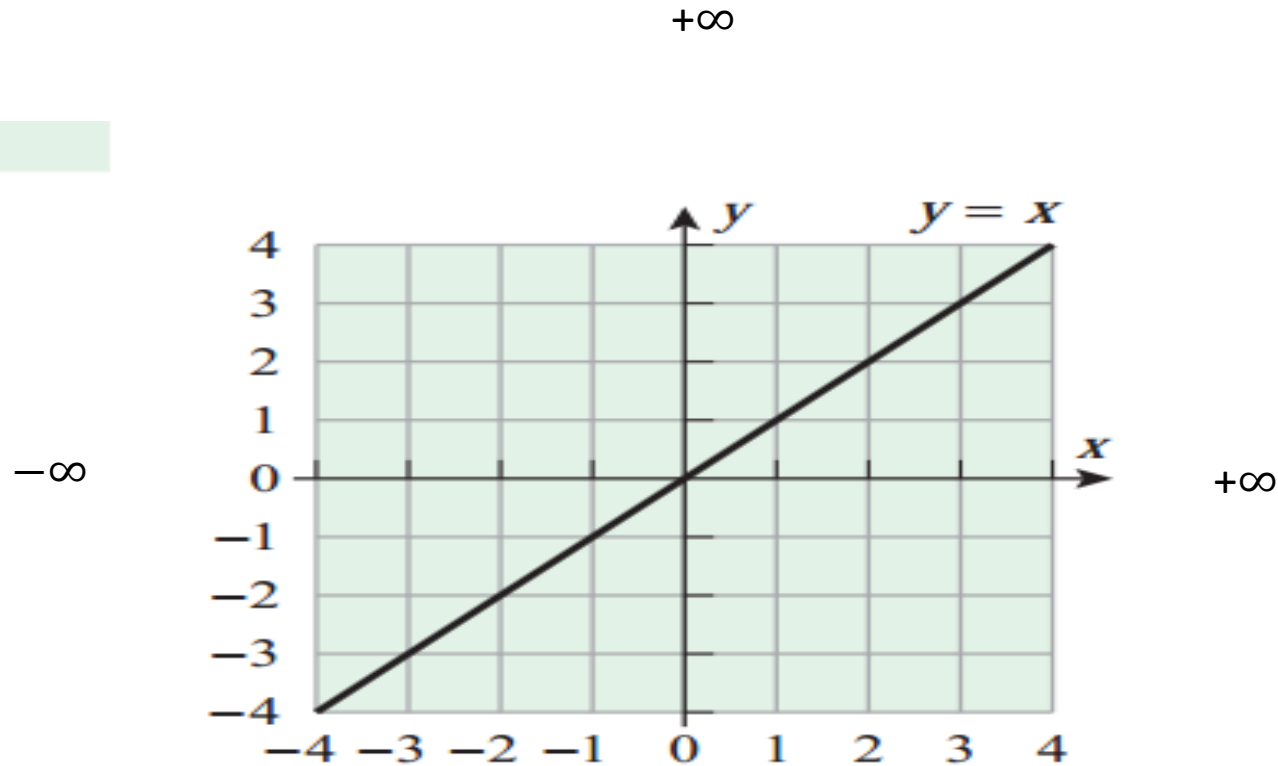
$$y = f(x),$$

then the set of all allowable inputs (x -values) is called the *domain* of f , and the set of outputs (y -values) that result when x varies over the domain is called the *range* of f .

Draw the graphs. Also find the domain and range of the functions:

- $f(x) = x$
- $f(x) = x^2$
- $g(x) = x^3$
- $h(x) = \sqrt[3]{x} = x^{(\frac{1}{3})}$
- $q(x) = \sqrt{x}$
- $f(x) = |x|$

Finding domain and range of $y = f(x) = x$
Domain is $(-\infty, +\infty)$, and Range is $(-\infty, +\infty)$

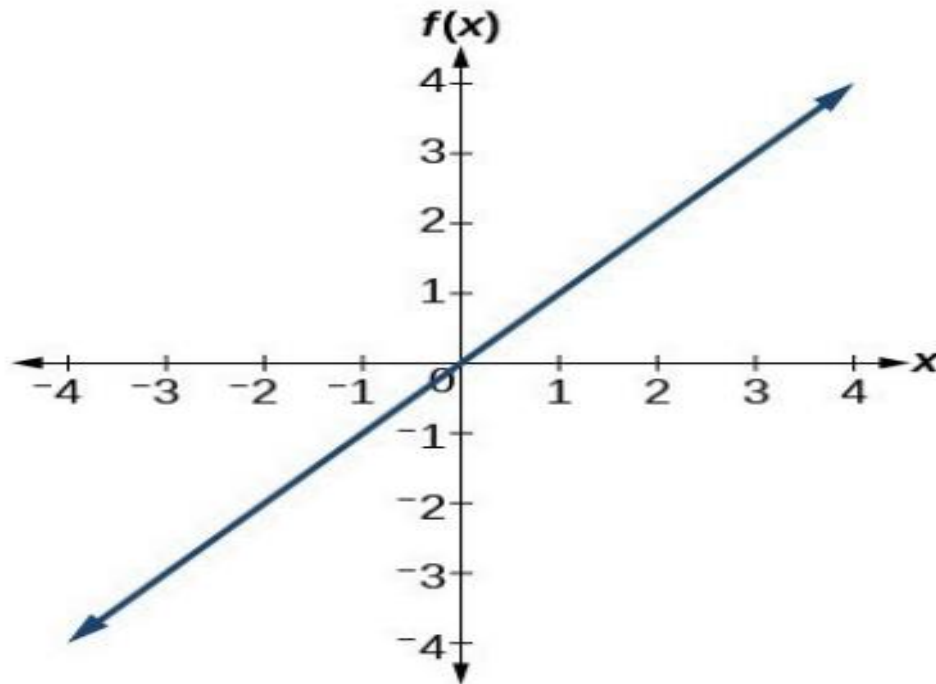


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Finding domain and range of $y = f(x) = x$

Input	Output
$x = -\infty$	$y = -\infty$
$x = -2$	$y = -2$
$x = -1$	$y = -1$
$x = 0$	$y = 0$
$x = 1$	$y = 1$
$x = 2$	$y = 2$
$x = +\infty$	$y = +\infty$

Graph of $y = f(x) = x$

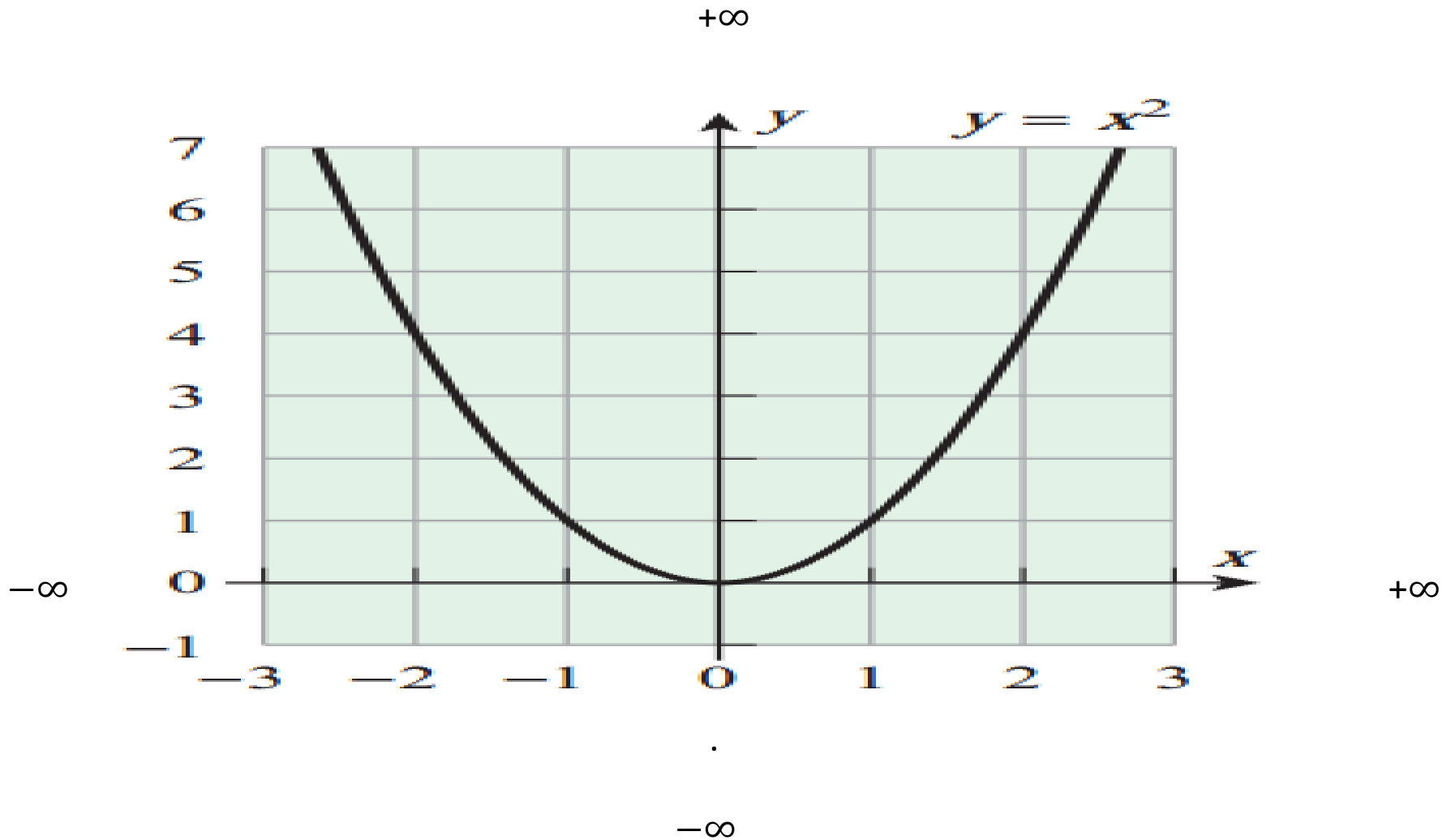


Domain: $(-\infty, \infty)$

Range: $(-\infty, \infty)$

For the **identity function** $f(x) = x$, there is no restriction on x . Both the domain and range are the set of all real numbers.

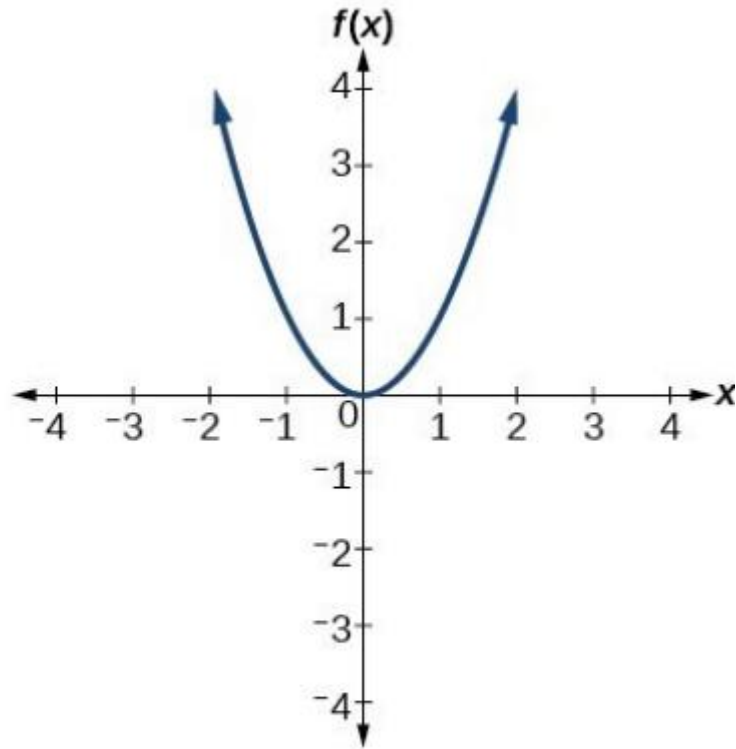
Finding domain and range of $y = f(x) = x^2$
Domain is $(-\infty, +\infty)$, and Range is $[0, +\infty)$



Finding domain and range of $y = f(x) = x^2$

Input	Output
$x = -\infty$	$y = +\infty$
$x = -2$	$y = 4$
$x = -1$	$y = 1$
$x = 0$	$y = 0$
$x = 1$	$y = 1$
$x = 2$	$y = 4$
$x = +\infty$	$y = +\infty$

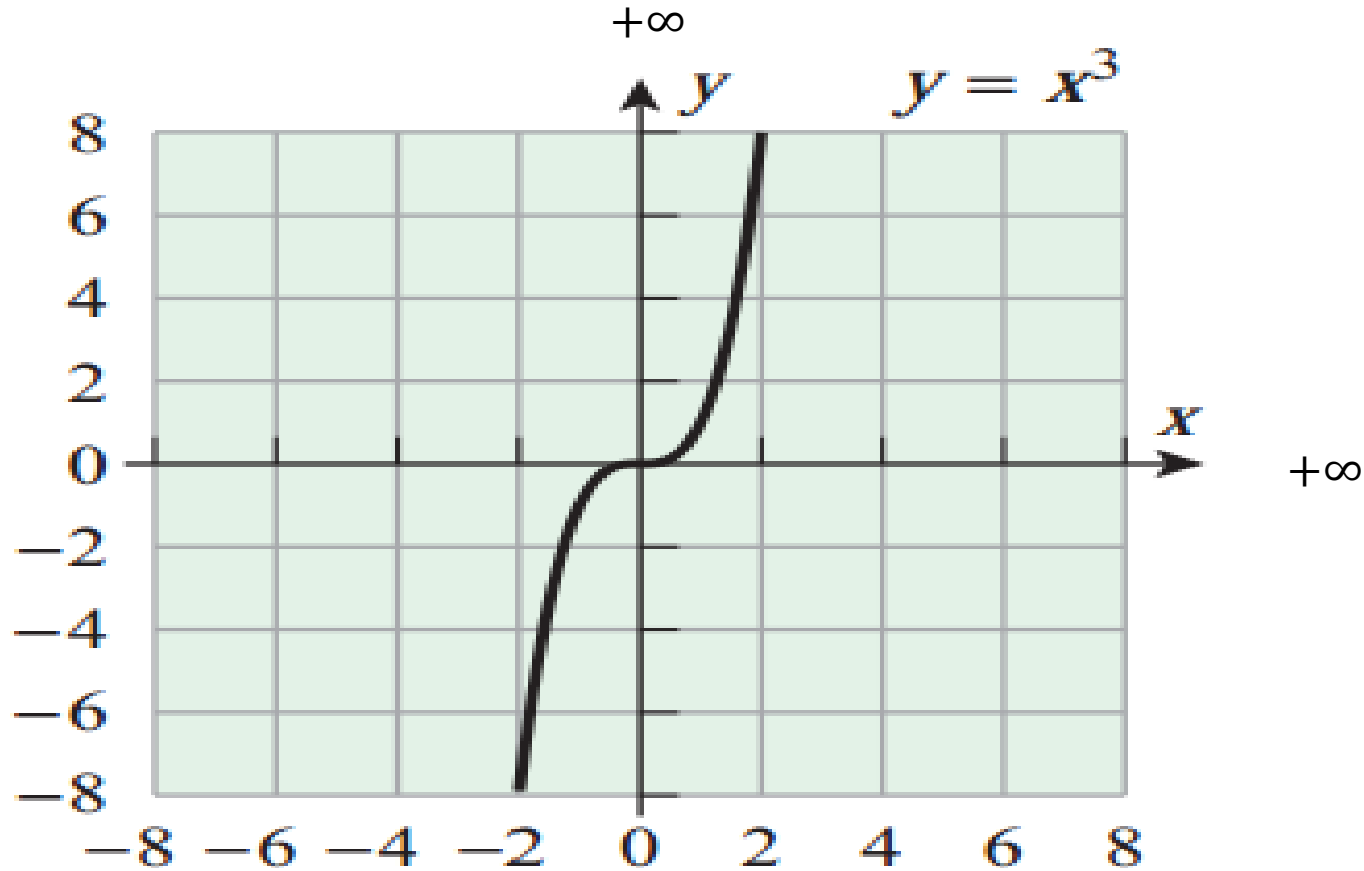
Graph of $y = f(x) = x^2$



Domain: $(-\infty, \infty)$

Range: $[0, \infty)$

Finding domain and range of $y = g(x) = x^3$
Domain is $(-\infty, +\infty)$, and Range is $(-\infty, +\infty)$



$-\infty$

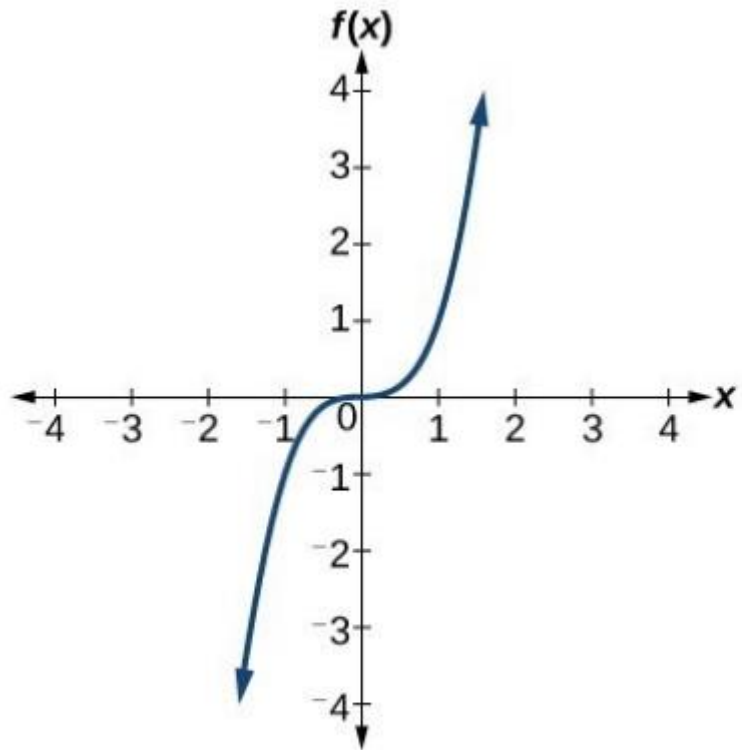
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Finding domain and range of

$$y = g(x) = x^3$$

Input	Output
$x = -\infty$	$y = -\infty$
$x = -2$	$y = -8$
$x = -1$	$y = -1$
$x = 0$	$y = 0$
$x = 1$	$y = 1$
$x = 2$	$y = 8$
$x = +\infty$	$y = +\infty$

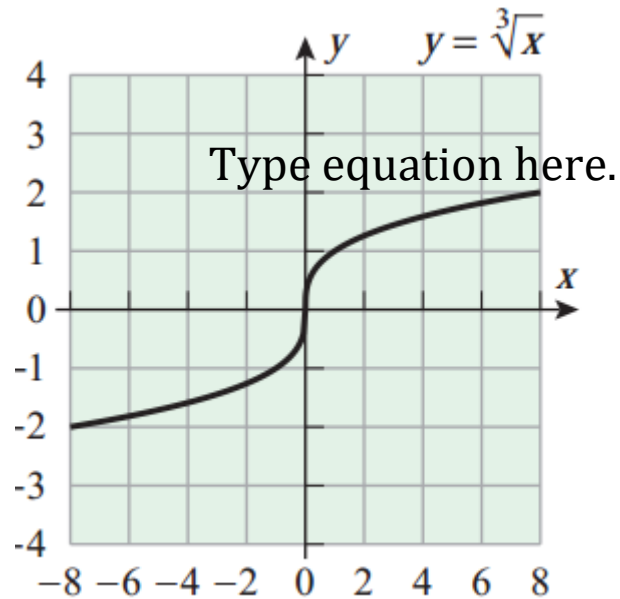
Graph of $y = g(x) = x^3$



Domain: $(-\infty, \infty)$

Range: $(-\infty, \infty)$

Finding domain and range of $y = h(x) = x^{\left(\frac{1}{3}\right)} = \sqrt[3]{x}$
Domain is $(-\infty, +\infty)$, and Range is $(-\infty, +\infty)$

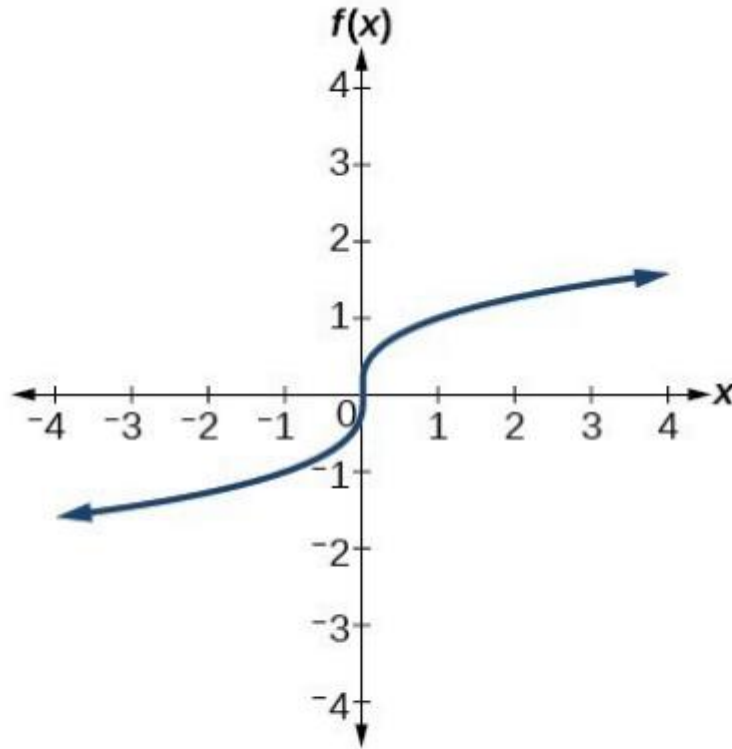


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Finding domain and range of $y = h(x) = x^{\left(\frac{1}{3}\right)} = \sqrt[3]{x}$

Input	Output
$x = -\infty$	$y = -\infty$
$x = -2$	$y = -1.259$
$x = -1$	$y = -1$
$x = 0$	$y = 0$
$x = 1$	$y = 1$
$x = 2$	$y = 1.259$
$x = +\infty$	$y = +\infty$

Graph of $y = h(x) = x^{\left(\frac{1}{3}\right)} = \sqrt[3]{x}$



Domain: $(-\infty, \infty)$

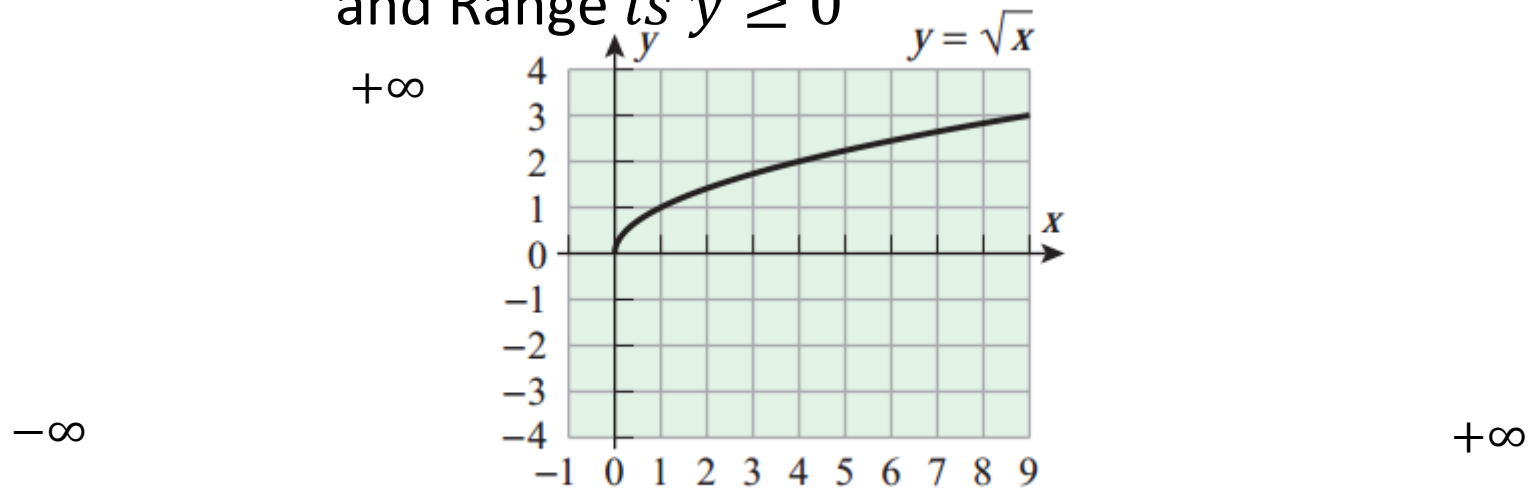
Range: $(-\infty, \infty)$

Finding domain and range of $y = f(x) = \sqrt{x}$.
Domain is $[0, +\infty)$, and Range is $[0, +\infty)$
or

Since f is defined if $x \geq 0$, so domain is $x \geq 0$.

If $x = 0$ then, $y = 0$

and Range is $y \geq 0$



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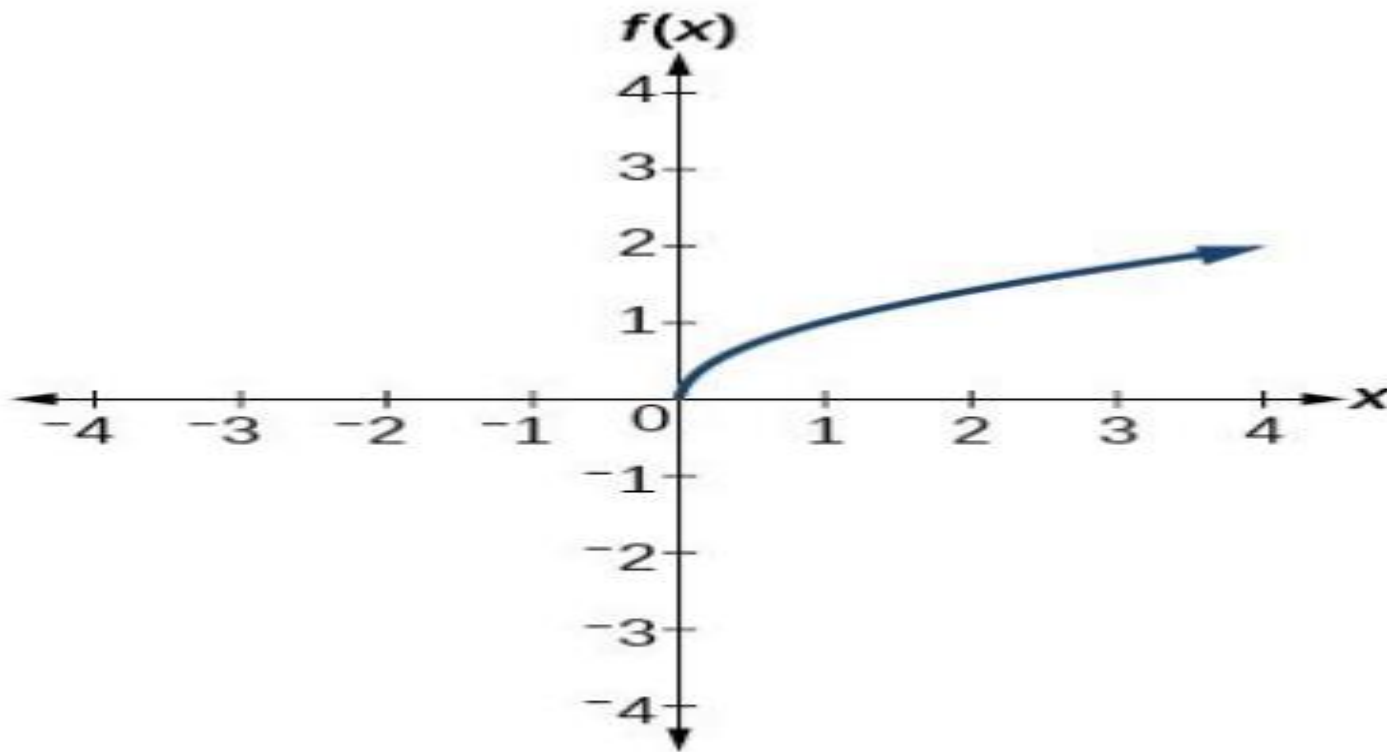
Finding domain and range of

$$y = f(x) = \sqrt{x}$$

Since f is defined if $x \geq 0$

Input	Output
$x = -\infty$	$y = \textit{undefined}$
$x = -2$	$y = \textit{undefined}$
$x = -1$	$y = \textit{undefined}$
$x = 0$	$y = 0$
$x = 1$	$y = 1$
$x = 2$	$y = 1.414$
$x = +\infty$	$y = +\infty$

Graph of $y = f(x) = \sqrt{x}$



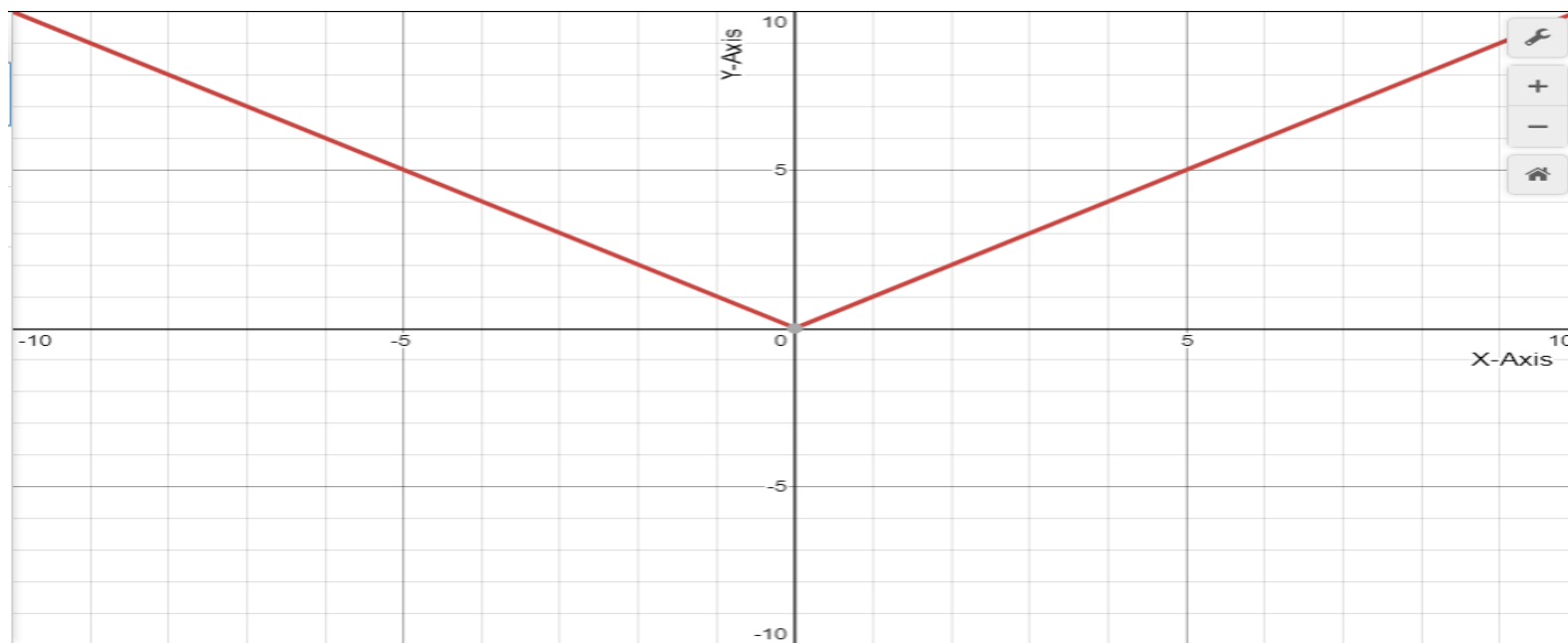
Domain: $[0, \infty)$
Range: $[0, \infty)$

THE ABSOLUTE VALUE FUNCTION

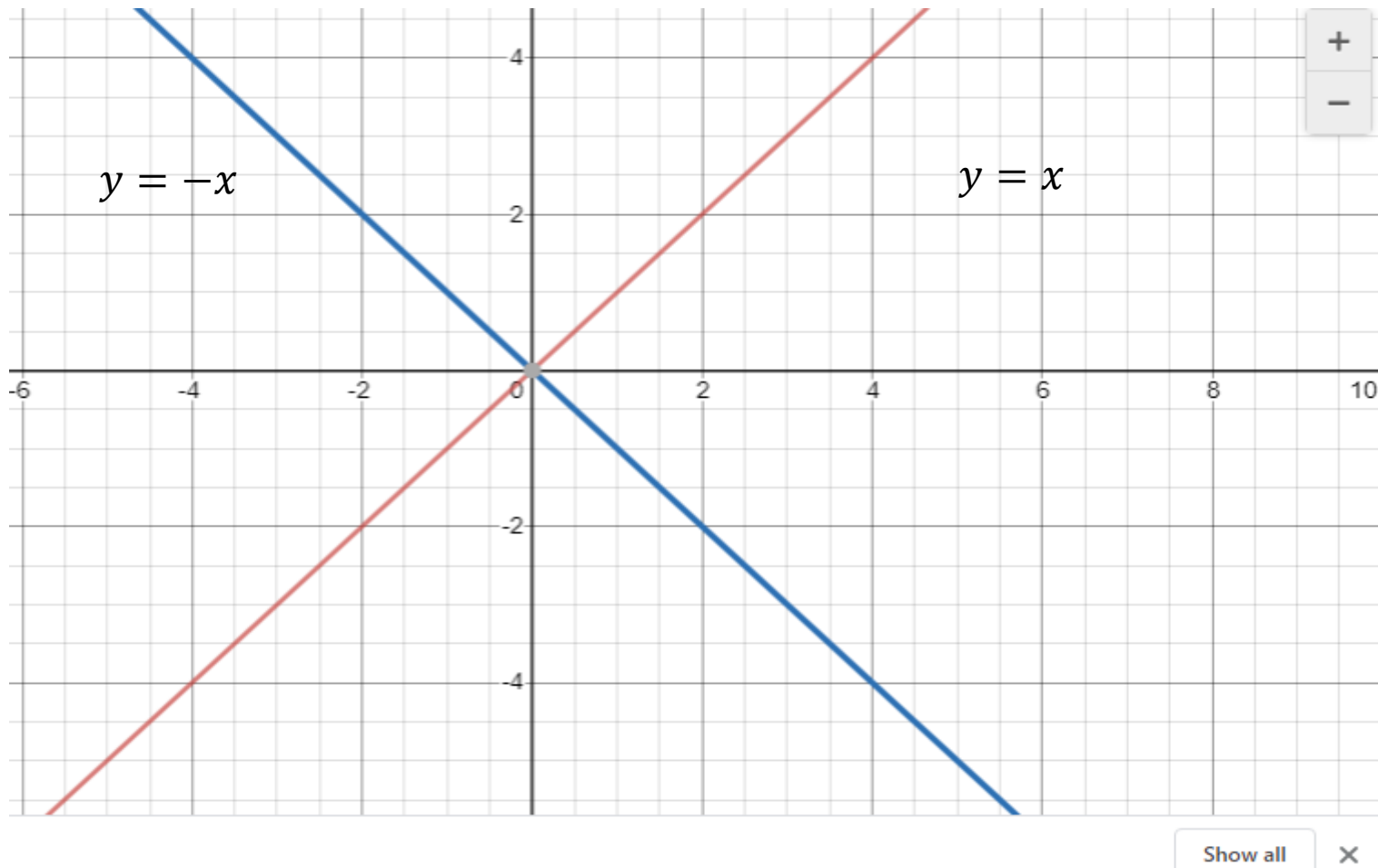
(Piece-wise defined function)

- The *absolute value* or *magnitude* of a real number x is defined by,

$$y = f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

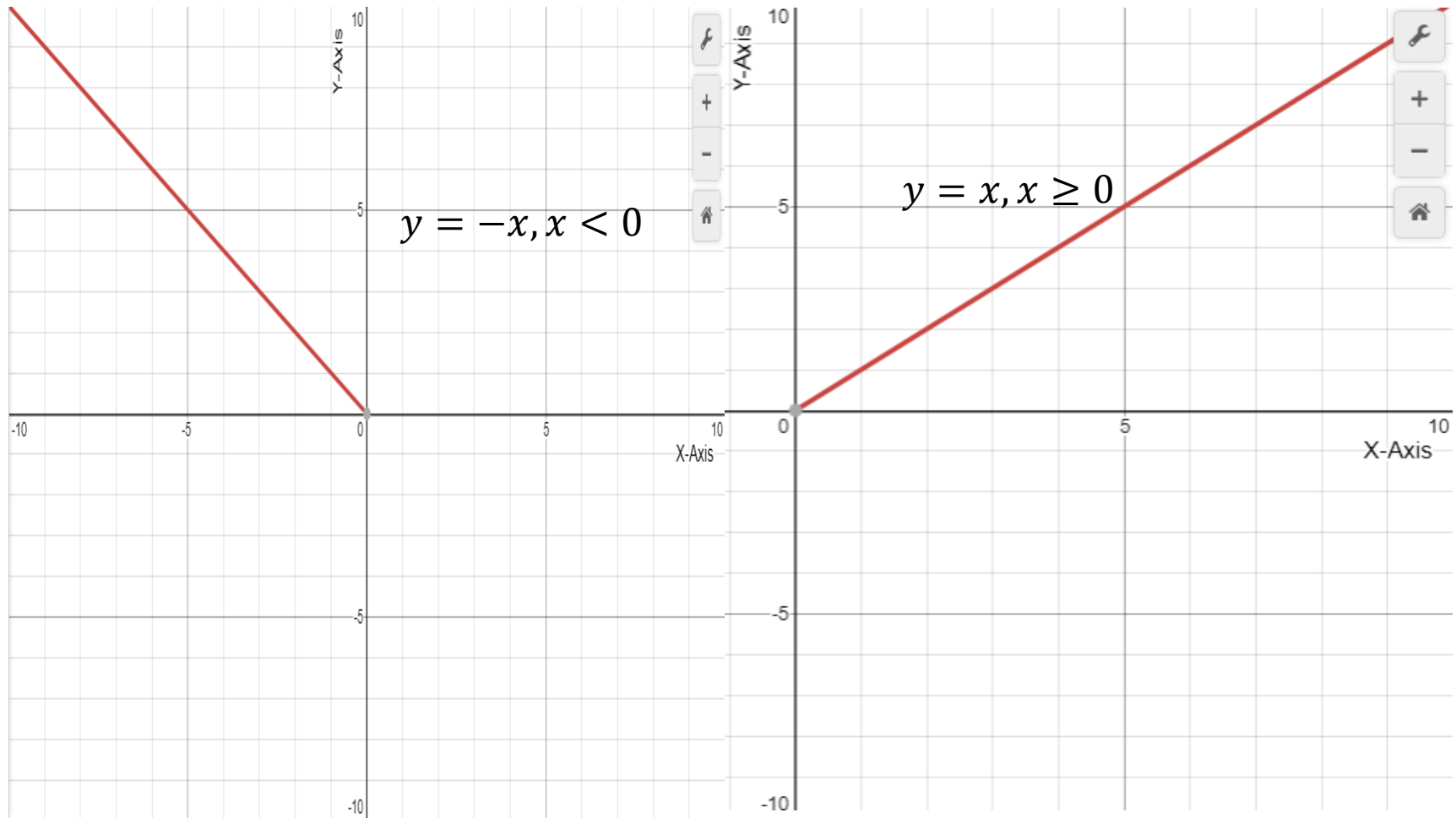


Graph of $y = x$, $y = -x$

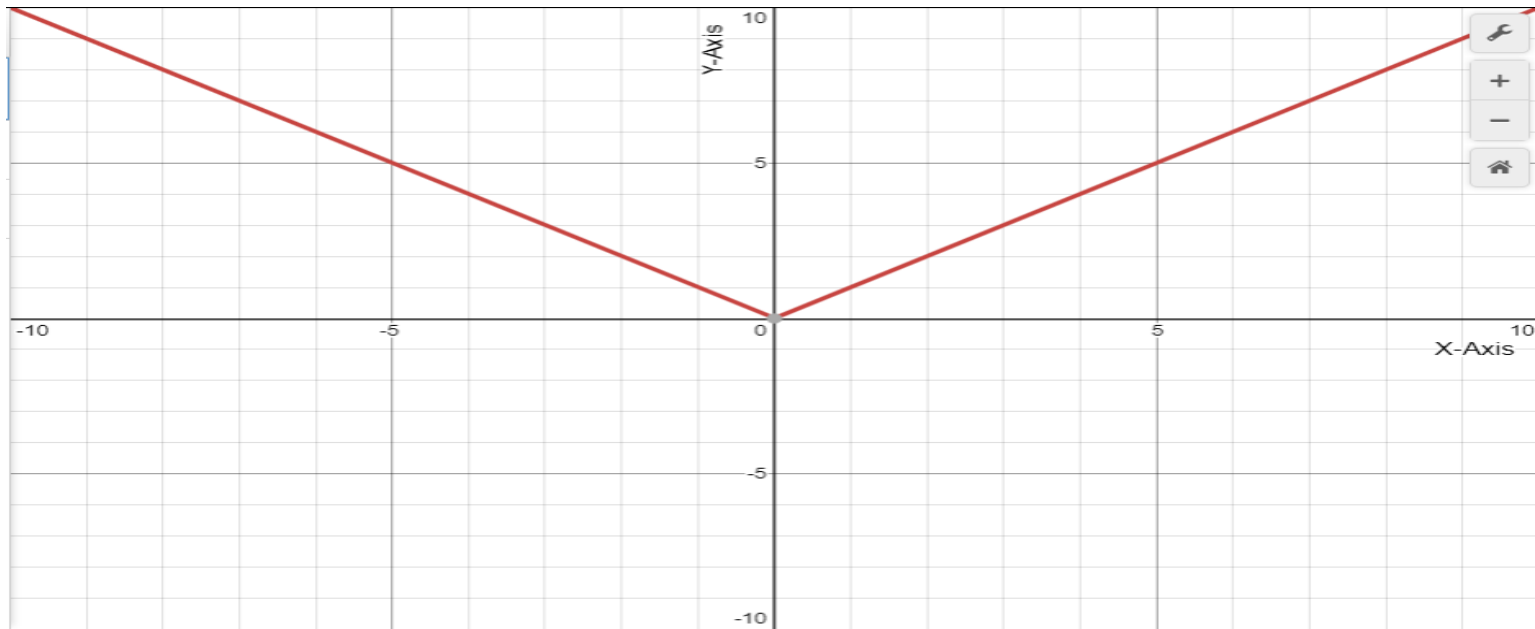


Graph of

$$y = x, x \geq 0; y = 0; x = 0 \text{ and } y = -x, x < 0$$

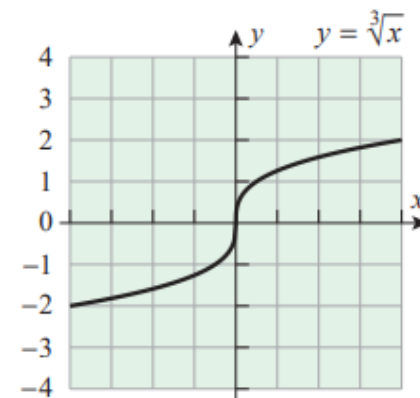
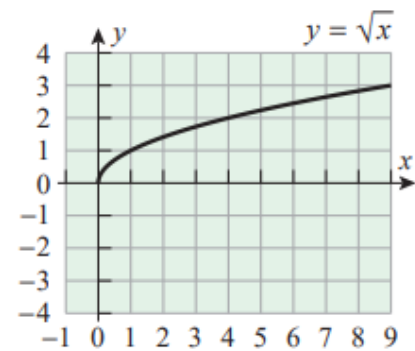
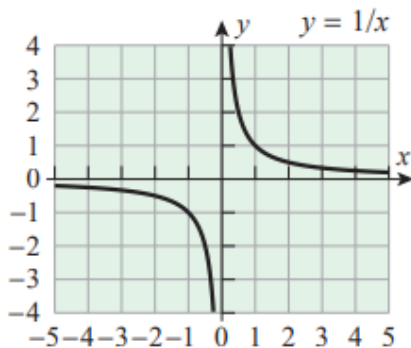
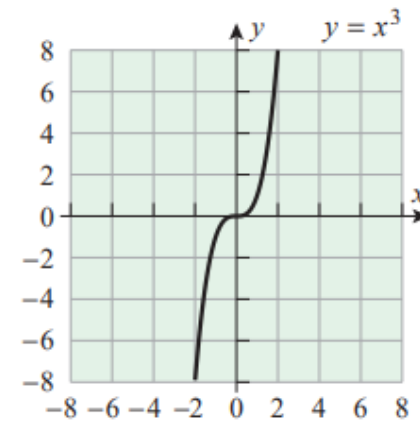
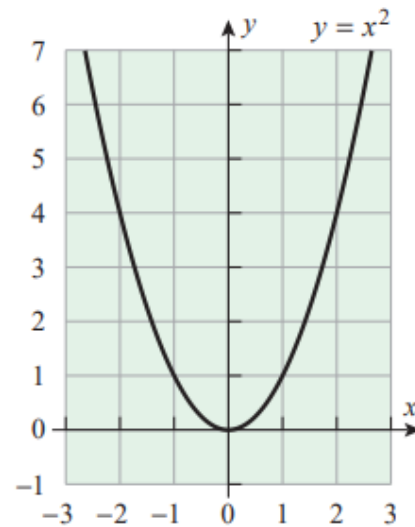
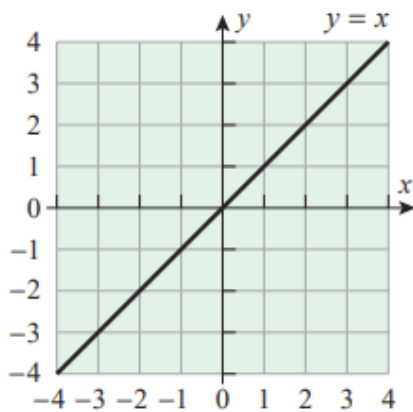


Graph of $y = |x|$



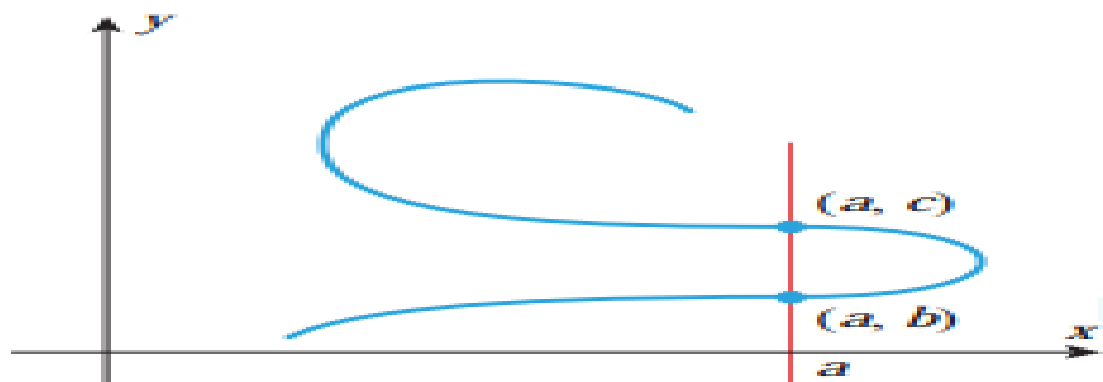
Domain is $\mathbb{R}$ or $(-\infty, +\infty)$ and range is $[0, +\infty)$

Graph of functions



Vertical line test

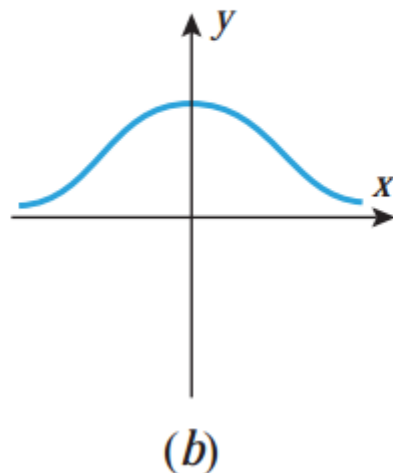
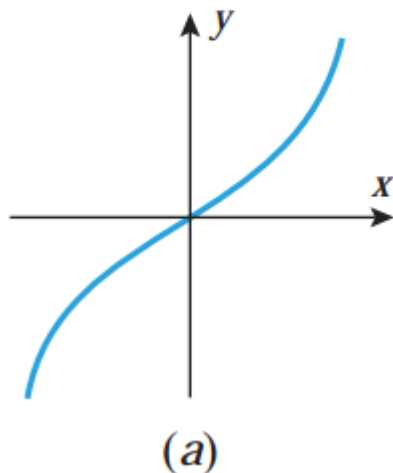
0.1.3 THE VERTICAL LINE TEST *A curve in the xy -plane is the graph of some function f if and only if no vertical line intersects the curve more than once.*



▲ **Figure 0.1.7** This curve cannot be the graph of a function.

Vertical line test

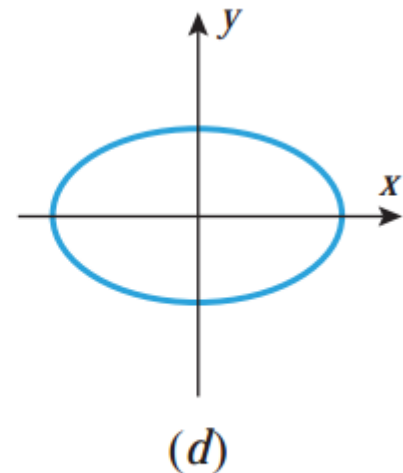
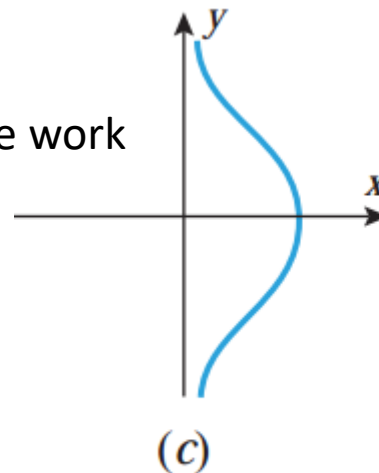
3. In each part of the accompanying figure, determine whether the graph defines y as a function of x .



Home
work

Home
work

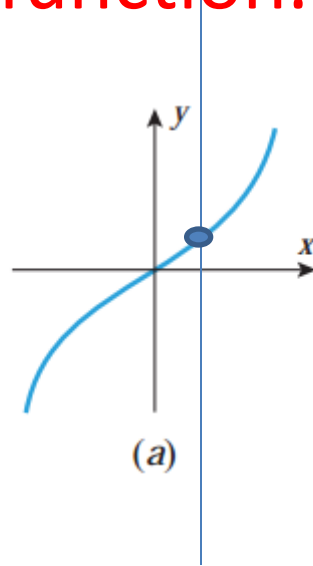
Determine whether
the given curve is
function or not.



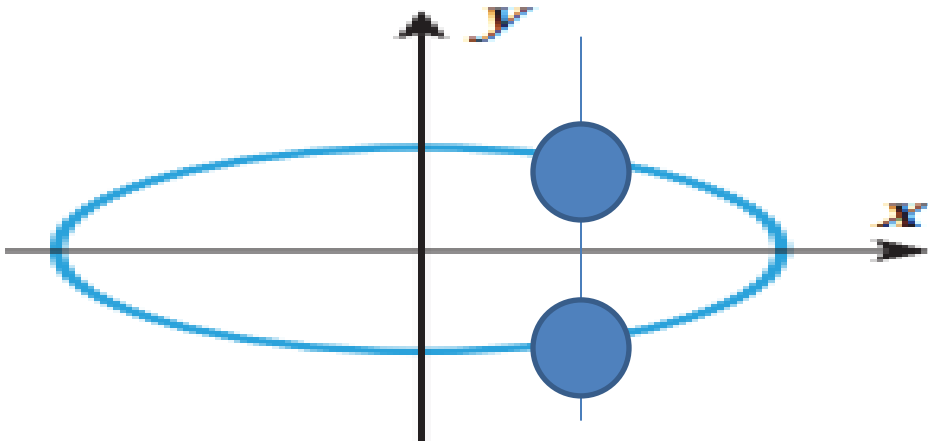
▲ Figure Ex-3

Solution: 3(a):

From Vertical line test we can say that, the vertical line cuts or intersects the graph only one point, so the given curve is a graph of function.



- Solution: 3(d):
- From Vertical line test we can say that, the vertical line cuts or intersects the graph more than one point, so the given curve is not a graph of function.



Function

7. Find $f(0)$, $f(2)$, $f(-2)$, $f(3)$, $f(\sqrt{2})$, and $f(3t)$.

$$(a) f(x) = 3x^2 - 2 \qquad (b) f(x) = \begin{cases} \frac{1}{x}, & x > 3 \\ 2x, & x \leq 3 \end{cases}$$

8. Find $g(3)$, $g(-1)$, $g(\pi)$, $g(-1.1)$, and $g(t^2 - 1)$.

$$(a) g(x) = \frac{x+1}{x-1} \qquad (b) g(x) = \begin{cases} \sqrt{x+1}, & x \geq 1 \\ 3, & x < 1 \end{cases}$$

Function

7. (a) $f(0) = 3(0)^2 - 2 = -2$; $f(2) = 3(2)^2 - 2 = 10$; $f(-2) = 3(-2)^2 - 2 = 10$; $f(3) = 3(3)^2 - 2 = 25$; $f(\sqrt{2}) = 3(\sqrt{2})^2 - 2 = 4$; $f(3t) = 3(3t)^2 - 2 = 27t^2 - 2$.

(b) $f(0) = 2(0) = 0$; $f(2) = 2(2) = 4$; $f(-2) = 2(-2) = -4$; $f(3) = 2(3) = 6$; $f(\sqrt{2}) = 2\sqrt{2}$; $f(3t) = 1/(3t)$ for $t > 1$ and $f(3t) = 6t$ for $t \leq 1$.

8. (a) $g(3) = \frac{3+1}{3-1} = 2$; $g(-1) = \frac{-1+1}{-1-1} = 0$; $g(\pi) = \frac{\pi+1}{\pi-1}$; $g(-1.1) = \frac{-1.1+1}{-1.1-1} = \frac{-0.1}{-2.1} = \frac{1}{21}$; $g(t^2 - 1) = \frac{t^2 - 1 + 1}{t^2 - 1 - 1} = \frac{t^2}{t^2 - 2}$.

(b) $g(3) = \sqrt{3+1} = 2$; $g(-1) = 3$; $g(\pi) = \sqrt{\pi+1}$; $g(-1.1) = 3$; $g(t^2 - 1) = 3$ if $t^2 < 2$ and $g(t^2 - 1) = \sqrt{t^2 - 1 + 1} = |t|$ if $t^2 \geq 2$.

Thank you